

Problem Set 1

March 2, 2026

- 1 Let $x[n] = \cos \frac{5\pi n}{7} + \cos \frac{\pi n}{5}$. Is $x[n]$ a periodic signal? If so, calculate its period.
- 2 By direct evaluation of the convolution sum, determine the unit step response ($x[n] = u[n]$) of an LTI system whose impulse response is

$$h[n] = a^{-n}u[-n], \quad 0 < a < 1.$$

- 3 Consider the linear constant-coefficient difference equation

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n-1].$$

Determine $y[n]$ for $n \geq 0$ when $x[n] = \delta[n]$ and $y[n] = 0, n < 0$.

- 4 A sequence has the DTFT

$$X(e^{j\omega}) = \frac{1 - a^2}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}, \quad |a| < 1.$$

- (a) Find the sequence $x[n]$.
- (b) Calculate $\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \cos(\omega) d\omega$.
- 5 (a) Determine the Fourier transform of the sequence

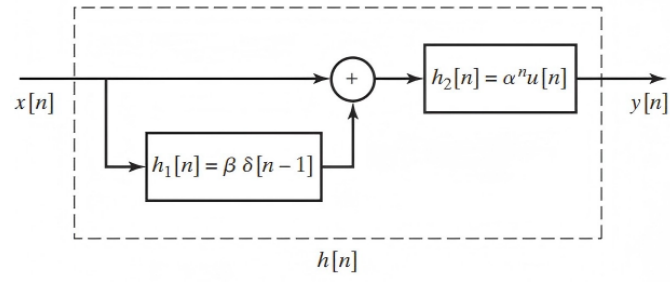
$$r[n] = \begin{cases} 1, & 0 \leq n \leq M, \\ 0, & \text{otherwise.} \end{cases}$$

- (b) Consider the sequence

$$w[n] = \begin{cases} \frac{1}{2} [1 - \cos(\frac{2\pi n}{M})], & 0 \leq n \leq M, \\ 0, & \text{otherwise.} \end{cases}$$

Sketch $w[n]$ and express $W(e^{j\omega})$, the Fourier transform of $w[n]$, in terms of $R(e^{j\omega})$, the Fourier transform of $r[n]$. (*Hint:* First express $w[n]$ in terms of $r[n]$ and the complex exponentials $e^{j(2\pi n/M)}$ and $e^{-j(2\pi n/M)}$.)

- (c) Sketch the magnitude of $R(e^{j\omega})$ and $W(e^{j\omega})$ for the case when $M = 4$.



6 Consider the system in above Figure.

- Determine the impulse response $h[n]$ of the overall system.
- Determine the frequency response of the overall system.
- Specify a difference equation that relates the output $y[n]$ to the input $x[n]$.
- Is this system causal? Under what condition would the system be stable?